

ZHE WANG

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Education

University of Illinois at Urbana-Champaign

Master of Science in Computer Science

- **Advisor:** Lingming Zhang

Aug 2024 – May 2026

Illinois, United States

Tsinghua University

Bachelor of Science in Mathematics and Physics

- **Advisor:** Zhiyuan Liu

Sep 2020 – Jun 2024

Beijing, China

Research Interests

- **LLMs for Code:** to develop LLMs to solve software engineering tasks through post-training via synthetic data.
- **Trustworthy LLMs:** to enhance trustworthiness, resilience and reliability of helpful-only LLMs against vulnerable code and malicious cyberactivity attacks.
- **LLM Applications:** to empower LLMs with reasoning, planning and collaboration capabilities through alignment training and agent-based systems.

Publications

- Chunqiu Steven Xia, **Zhe Wang**, Yan Yang, Yuxiang Wei, Lingming Zhang. “*Live-SWE-agent: Can Software Engineering Agents Self-Evolve on the Fly?*”, arxiv preprint. [\[Paper\]](#) [\[Code\]](#) [\[Leaderboard\]](#)
- Jiawei Liu*, Nirav Diwan*, **Zhe Wang***, Haoyu Zhai, Xiaona Zhou, Kiet A. Nguyen, Tianjiao Yu, Muntasir Wahed, Yinlin Deng, Hadjer Benkraouda, Yuxiang Wei, Lingming Zhang, Ismini Lourentzou, Gang Wang. “*PurpCode: Reasoning for Safer Code Generation*”, NeurIPS 2025. [\[Paper\]](#) [\[Code\]](#) [\[Dataset\]](#)
- Kunlun Zhu*, Hongyi Du*, Zhaochen Hong*, Xiaocheng Yang*, Shuyi Guo*, **Zhe Wang***, Zhenhailong Wang, Cheng Qian, Xiangru Tang, Heng Ji, Jiaxuan You. “*MultiAgentBench: Evaluating the Collaboration and Competition of LLM Agents*”, ACL main 2025. [\[Paper\]](#) [\[Code\]](#)
- Yuxiang Wei, **Zhe Wang**, Jiawei Liu, Yifeng Ding, Lingming Zhang. “*Magocoder: Empowering code generation with oss-instruct*”, ICML 2024. [\[Paper\]](#) [\[Code\]](#) [\[Dataset\]](#)

Research Experiences

Live-SWE-Agent: Can Software Engineering Agents Self-Evolve on the Fly?

Supervised by Prof. Lingming Zhang at UIUC

Sept 2025 – Nov 2025

- Introduced **Live-SWE-Agent**, the first live software agent that can autonomously and continuously evolve itself on-the-fly during runtime when solving real-world software issues.
- Live-SWE-Agent outperformed all existing OSS software agents and approaching the performance of the best proprietary solution on SWE-bench Verified and SWE-Bench Pro without any offline training or heavy workflows.

PurpCode: Reasoning for Safer Code Generation

Supervised by Prof. Gang Wang, Prof. Lourentzou, Prof. Lingming Zhang at UIUC

Dec 2024 – Jul 2025

- Introduced **PurpCode**, the first cybersafety reasoning model that generates secure code and defends against malicious cyber activities, outperforming frontier models in secure code generation while maintaining utility.
- Designed a novel training pipeline with (1)**Data Curation**, (2)**Rule Learning** via SFT on self-aligned safety reasoning trajectories, and (3) **Reinforcement Learning** with multi-objective rewards.
- **Won 1st place in Amazon Nova AI Challenge 2025.**

MultiAgentBench: Evaluating the Collaboration and Competition of LLM Agents

Supervised by Prof. Jiaxuan You and Prof. Heng Ji at UIUC

Nov 2024 – Feb 2025

- Introduced **MultiAgentBench**, a comprehensive benchmark, and the **MARBLE** framework for evaluating LLM-based multi-agent systems across 6 different collaborative and competitive scenarios.
- Developed innovative evaluation metrics including milestone-based KPIs, planning/communication scores, and competition metrics, offering insights through emergent social behaviors toward AGI-level collaboration.
- Project accepted at ACL main 2025. Code and dataset released at Github.

Magocoder: Empowering Code Generation with OSS-Instruct

Supervised by Prof. Lingming Zhang at UIUC

Oct 2023 – Dec 2023

- Proposed **OSS-Instruct**, a new approach for high-quality data synthesis with open-source code snippets, adopted by *Meta Llama 3.1*, *Google CodeGemma*, and *IBM Granite*.
- Introduced **Magocoder**, a series of fully open-source coding LLMs trained with OSS-Instruct, outperforming all evaluated LLMs of ≤ 16 B parameters with only 7B parameters.
- **Achieved 2k+ Github stars, featured on Github trending, and received 393k+ downloads.**

Technical Skills

Languages: Mandarin (Native), English (TOEFL Best Score: 105 (**Speaking 25**)), Japanese Beginner

Programming Languages: Python, Java, C, HTML/CSS, JavaScript, SQL

Technologies/Frameworks: Linux, GitHub, Huggingface, Pytorch